MO-HA-II

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① Use Branch & Bound method to Maximize ~~$Z = 5x_1 + 3x_2$~~

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S.B => $4x_1 + 5x_2 \le 6$

$x_1 + 3x_2 \le 8$ x_1, x_2 are non-ve integers

$7x_2 \le 8$

sol)

Given, $Z = 5x_1 + 3x_2$

SB => $4x_1 + 5x_2 \le 6$

$x_1 + 3x_2 \le 8$

$7x_2 \le 8$

$x_1, x_2 \ge 0$

From 3rd eq $7x_2 \le 8$

$x_2 \le 1.14$.

$x_2 \approx 0$ (or) 1.

check feasible solutions:-

Case-1 :- $x_2 = 0$

$4x_1 \le 6$

$x_1 \le 1.5$

$x_1 = 0, 1$

z values :-

$(0,0) \rightarrow z = 0$

$(1,0) \rightarrow z = 5$

Case-2 :-

$4x_1 + 5 \le 6 \Rightarrow x_1 \le 0.25$

$x_1 = 0$

z value :

$(0,1) \rightarrow z = 3$

choosing maximum :- $x_1 = 1$, $x_2 = 0$

$Z_{max} = 5(1) + 3(0)$.

$= 5$.

2. Use Newton's method to find minimum value of $f(x) = x^2 - 6x + 20$ with initial guess $x_0 = 0$.

Sol :- $f(x) = x^2 - 6x + 20$

$f'(x) = 2x - 6$

$f''(x) = 2$

Formula :-

$$x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)}$$

Given $x_0 = 0$:-

$$x_1 = 0 - \frac{-6}{2} = 3$$

check minimum : -

$f''(x) = 2 > 0 \Rightarrow$ minimum

$f(3) = 9 - 18 + 20$

$= 11$

$\therefore x = 3, f(x) = 11$

3) Solve the objective function by Gomery's cut plane method. Objective function Max $Z = x_1 + 4x_2$

sub to : $6x_1 + 4x_2 \le 24$

$2x_1 + 2x_2 \le 6$

$x_1, x_2 \ge 0$

Sol :-

$Z = x_1 + 4x_2$

sub to :- $6x_1 + 4x_2 \le 24$

$2x_1 + 2x_2 \le 6$

solving :- $x_1 + x_2 \le 3$

Corner points :-

(0, 0)

(0, 3) ✓

(4, 0)

check $(0, 3)$

$$z = 0 + 4(3)$$

$$z = 0 + 12$$

$$\underline{\underline{z = 12}}$$

$\therefore x_1 = 0, x_2 = 3, z_{max} = 12$. Since solution is integer no cut plane needed.

4) Algorithms steps involved in Newton's method for optimization. with an example.

Sol:- STEPS:-

1) Choose initial guess x_0
2) Compute first derivative $f'(x)$
3) Compute second derivative $f''(x)$.
4) Apply formula:- $x_{n+1} = x_n - \dfrac{f'(x_n)}{f''(x_n)}$
5) Repeat until convergence :- $|x_{n+1} - x_n| \leq 0$
6) check:-

$$\text{if } f''(x) > 0 \rightarrow \min$$
$$f''(x) < 0 \rightarrow \max$$

Ex:- $f(x) = x^2 - 4x + 5$

$f'(x) = 2x - 4$, $f''(x) = 2$

$$x_1 = 0 - \frac{-4}{2} = 2$$

Min at $x = 2$.

5) Steps involved in solving an integer programming problem using Branch & Bound method.

Sol:) STEPS :-

1. Ignore integer constraints and solve LP.

2. check solution:-

if integer → optimal solution

if not → proceed to branching

3. Branching:-

split the problem into subproblems:-

$x \le |value|$ and $x \ge |value|$

4. Bounding:-

- Calculating upper/lower bounds for each branch.

5. Prune Branches:-

- If bound is worse than current best → discard.

6. Repeat:-

Continue unit all branches are explored.

7. choose Best feasible integer solution.

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